



VMware vSphere Infrastructure Deployment Project

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1) Project Description

This project aims to design and implement a fully functional VMware vSphere environment in a controlled lab setup using three physical servers connected through a local wireless network. The objective is to simulate an enterprise-level virtualization infrastructure that demonstrates core VMware technologies, including ESXi, vCenter Server, shared storage, and advanced features such as vMotion, High Availability (HA), Distributed Resource Scheduler (DRS), and Fault Tolerance (FT).

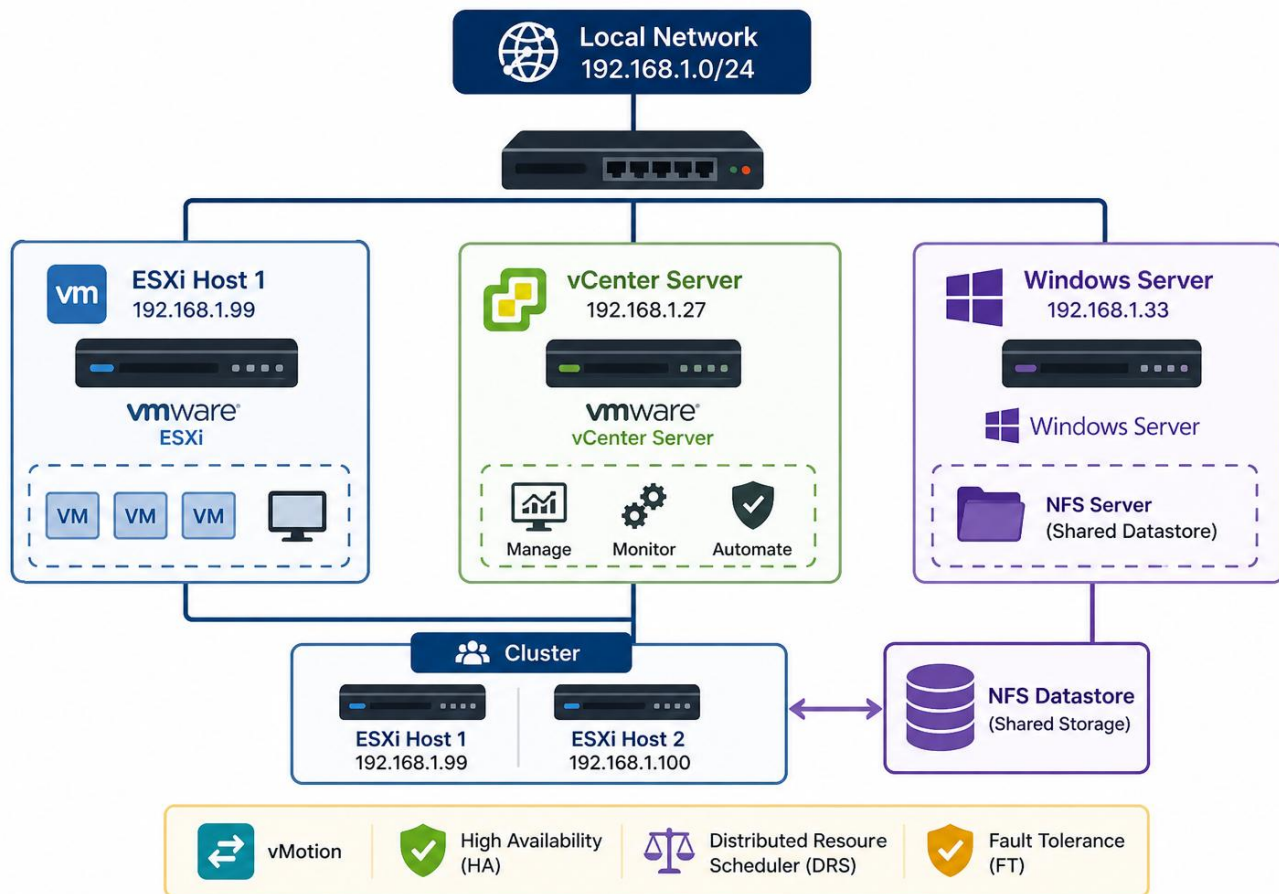
In this setup, two servers are configured as ESXi hypervisors, each hosting multiple virtual machines and participating in a centralized cluster managed by vCenter Server. The vCenter Server is deployed as a virtual appliance on one of the ESXi hosts, providing centralized management and monitoring of the entire virtual infrastructure. The third server runs Windows Server and is configured as an NFS shared datastore, offering centralized storage accessible to both ESXi hosts for virtual machine deployment, live migration, and resilience testing.

The network architecture is built using standard vSwitches with dedicated port groups for Management, vMotion, Storage, and VM Traffic, ensuring proper segmentation and optimized performance. Additionally, a Content Library is implemented to store ISO images and VM templates, enabling efficient provisioning and standardized deployment of virtual machines.

Virtual machines are deployed from the Content Library, and essential operations such as cloning, snapshot management, and template creation are performed to demonstrate full virtual machine lifecycle management.

By configuring HA and DRS within the cluster, the environment ensures automatic workload balancing and high availability in the event of host failure. Furthermore, Fault Tolerance is enabled for selected virtual machines to provide continuous availability with zero downtime.

Overall, this project provides a comprehensive hands-on implementation of VMware vSphere core functionalities, effectively bridging theoretical knowledge with practical enterprise virtualization experience and establishing a strong foundation for modern datacenter environments.



2) Setup ESXi and vCenter Server

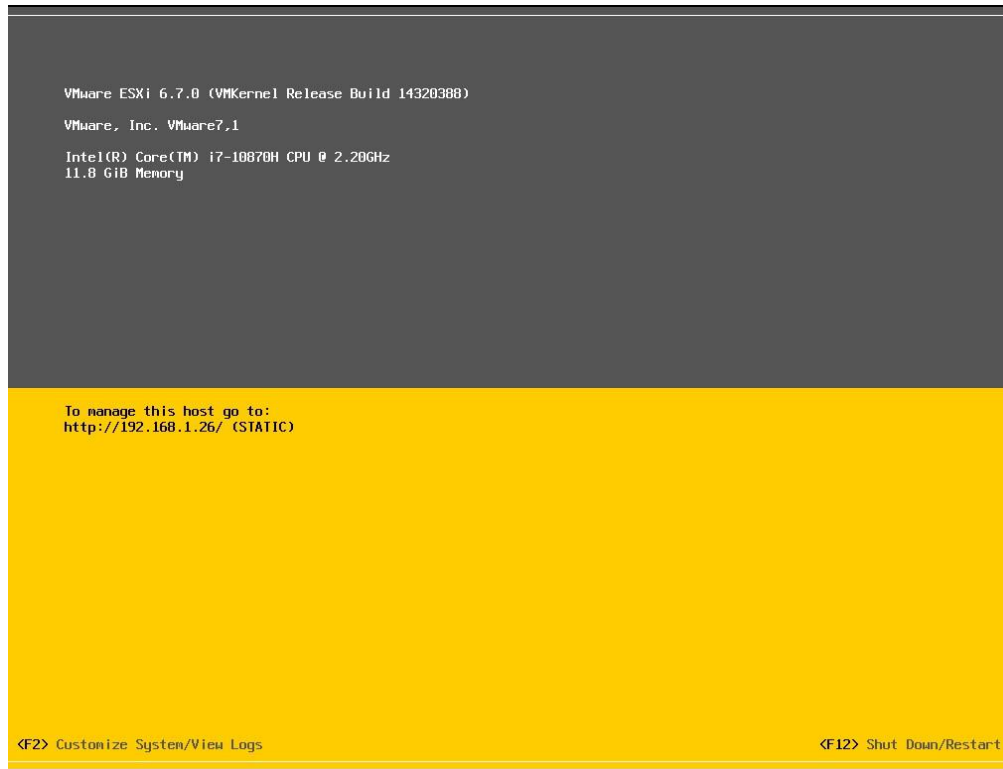
2.1) ESXI

ESXi is the name of the VMware vSphere hypervisor, which is a bare-metal hypervisor that runs directly on top of physical hardware, and abstracts the physical hardware resources to create Virtual Machines (VMs). ESXi version 8 is the latest version of the hypervisor. It is a type of server virtualization software that enables you to run multiple VMs on a single physical server, allowing you to consolidate your

workloads onto fewer servers and reduce your hardware and maintenance costs.

2.1.1) Deployment

In our Environment, we use ESXI version 6.7



2.2) vCenter Server

The purpose of the VCSA is to provide centralized management and automation of vSphere environments. It acts as a centralized management platform for vSphere environments, allowing administrators to manage and provision virtual resources such as VMs, networks, and storage.

By using VCSA, IT administrators can manage their vSphere infrastructure from a single console, increasing efficiency and reducing errors. It also provides advanced features such as automation, and it can be integrated with other tools such as vRealize Automation, vRealize Operations, and vRealize Log Insight, to provide a comprehensive management solution.

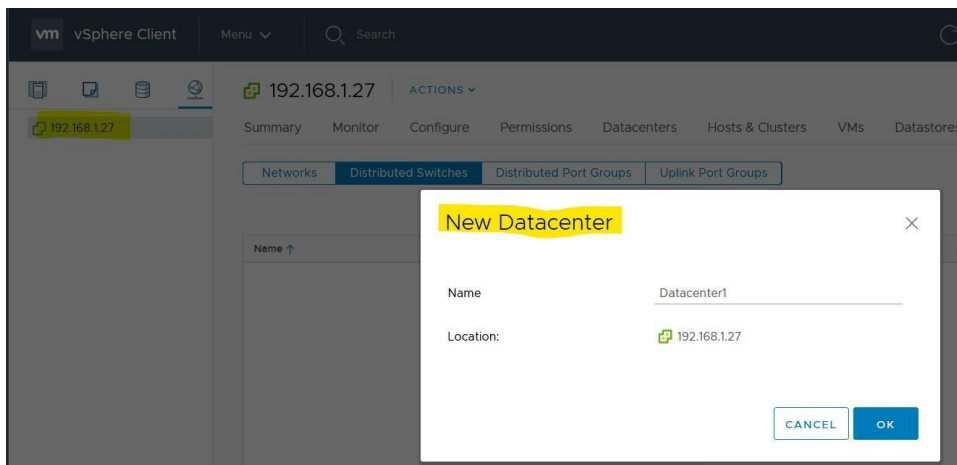
2.2.1) Deployment:

In our Environment, we use vCenter version 6.7

2.2.2) Configuration:

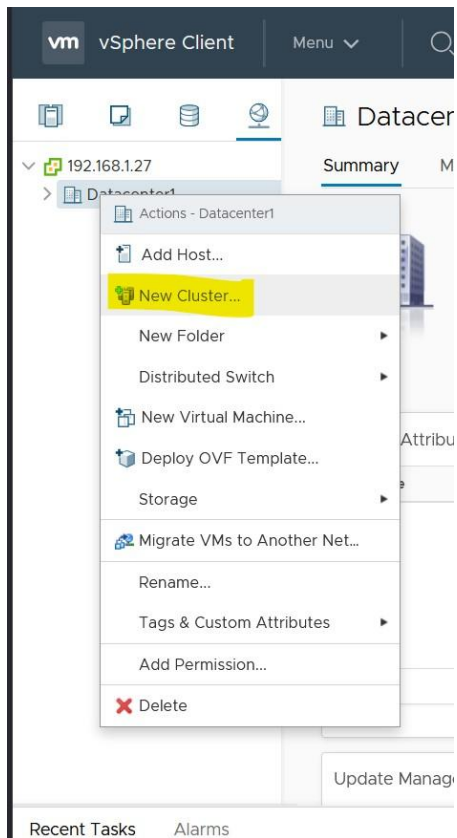
We configure a datacenter in vCenter to logically group and manage all hosts, clusters, storage, and networks in one place. A cluster is then created within the datacenter to pool ESXi hosts together, enabling advanced features like HA, DRS, and vMotion for resource sharing and high availability.

1) Create Datacenter



2) Create Cluster

- We enable HA to ensure VMs automatically restart on another host during failures, and DRS to balance workloads across hosts for optimal performance and resource utilization (to make VM happy).**

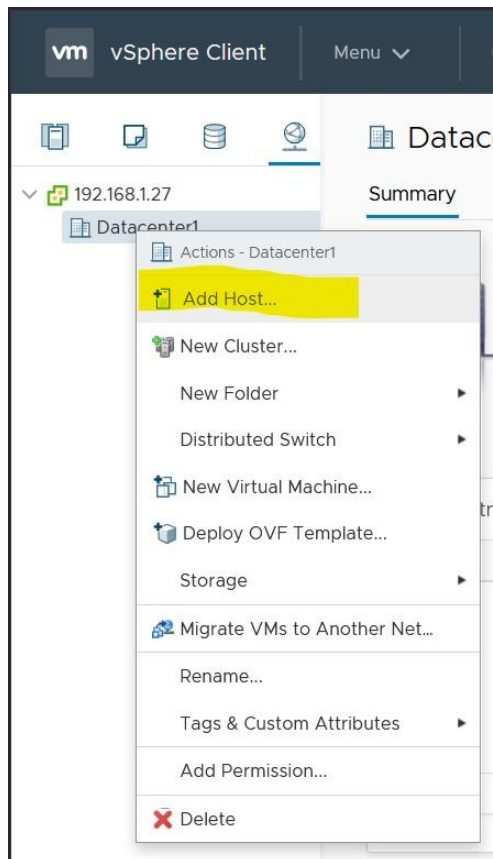


New Cluster | Datacenter1

Name	Project_cluster
Location	Datacenter1
> DRS	☒ Turn ON
> vSphere HA	☒ Turn ON
> EVC	Disable

3) Add ESXI Host to Cluster

- We add ESXi hosts to the cluster so they can share resources, participate in HA/DRS.



Add Host

- 1 Name and location
- 2 Connection settings
- 3 Host summary
- 4 Assign license
- 5 Lockdown mode
- 6 VM location
- 7 Ready to complete

Name and location

Enter the name or IP address of the host to add to vCenter Server.

Host name or IP address:	192.168.1.26
Location:	Datacenter1

Add Host

- ✓ 1 Name and location
- ✓ 2 Connection settings
- 3 Host summary
- 4 Assign license
- 5 Lockdown mode
- 6 VM location
- 7 Ready to complete

Connection settings

Enter the host connection details

User name:	root
Password:	

- Lockdown mode is a way to access the ESXI

When enabled, administrators cannot log in directly to the ESXi host (via SSH or DCUI) and must manage it only through vCenter Server. This prevents unauthorized changes and enforces centralized management, which is especially important in production environments.

Add Host

✓ 1 Name and location

✓ 2 Connection settings

✓ 3 Host summary

✓ 4 Assign license

5 Lockdown mode

6 VM location

7 Ready to complete

Lockdown mode
Specify whether to enable lockdown mode on the host

When enabled, lockdown mode prevents remote users from logging directly into this host. The host will only be accessible through local console or an authorized centralized management application.

If you are unsure what to do, leave lockdown mode disabled. You can configure lockdown mode later by editing Security Profile in host settings.

☒ Disabled

☐ Normal
The host is accessible only through the local console or vCenter Server.

☐ Strict
The host is accessible only through vCenter Server. The Direct Console UI service is stopped.

Add Host

✓ 1 Name and location

✓ 2 Connection settings

✓ 3 Host summary

✓ 4 Assign license

✓ 5 Lockdown mode

✓ 6 VM location

7 Ready to complete

Ready to complete
Click Finish to add the host

Name	192.168.1.26
Location	Datacenter1
Version	VMware ESXi 6.7.0 build-14320388
License	Evaluation License
Networks	VM Network
Datastores	datastore1
Lockdown mode	Disabled
VM location	Datacenter1

CANCEL

BACK

FINISH

3) Network

Creating Standard Switches in vSphere is important because it allows for the management of virtual networking in a vSphere environment. Standard Switches provide a virtual representation of a physical network switch and allow virtual machines to connect to virtual

networks and to physical networks through the use of virtual NICs (vNICs).

3.1) Network Environment Structure:

Management Network → Used by vCenter to connect to ESXi hosts and perform administrative tasks such as monitoring, configuration, and cluster management.

VM Network → Provides network connectivity for the virtual machines to communicate with each other and with external systems.

Storage Network → Dedicated network for NFS datastore traffic, ensuring reliable and isolated access between ESXi hosts and shared storage.

vMotion Network → Used exclusively for live migration (vMotion) traffic, allowing VMs to move between ESXi hosts without downtime.

- **No Virtual Resource without Physical Resource.**
- **In our Environment we use two physical NIC to implement two standard switches.**
- **We apply virtual switches (vSwitches) one for management and VMs port groups, and another switch for storage and migration.**

The screenshot displays the vSphere vSwitch configuration interface. At the top, the title bar reads 'vSwitch0'. Below it, a toolbar contains 'Add uplink', 'Edit settings', 'Refresh', and 'Actions' buttons. A summary section shows 'vSwitch0' as a 'Standard vSwitch' with '2' port groups and '1' uplink. The main area is divided into two panels: 'vSwitch Details' and 'vSwitch topology'.

vSwitch Details

MTU	1500
Ports	2560 (2549 available)
Link discovery	Listen / Cisco discovery protocol (CDP)
Attached VMs	1 (0 active)
Beacon interval	1

NIC teaming policy

Notify switches	Yes
Policy	Route based on originating port ID
Reverse policy	Yes

vSwitch topology

This panel shows the network topology. On the left, two network stacks are visible: 'VM Network' (VLAN ID: 0) with one virtual machine 'vsphere', and 'Management Network' (VLAN ID: 0) with one VMkernel port 'vmk0' at IP 192.168.1.26. These are connected to a central switch icon, which is in turn connected to a 'Physical adapters' block on the right, specifically 'vmnic0' at 10000 Mbps in Full Duplex mode.

vSwitch1

Add uplink
Edit settings
Refresh
Actions

vSwitch1
Type: Standard vSwitch
Port groups: 2
Uplinks: 1

vSwitch Details

MTU	1500
Ports	2560 (2549 available)
Link discovery	Listen / Cisco discovery protocol (CDP)
Attached VMs	0 (0 active)
Beacon interval	1

NIC teaming policy

Notify switches	Yes
Policy	Route based on originating port ID
Reverse policy	Yes

vSwitch topology

Physical adapters
vmnic1, 10000 Mbps, Full

3.2) Deployment

192.168.1.26

ACTIONS

Summary
Monitor
Configure
Permissions
VMs
Resource Pools
Datastores
Networks
Updates

192.168.1.27

Datacenter1

Project_cluster

192.168.1.26

vsphere

Storage

Storage Adapters

Storage Devices

Host Cache Configur...

Protocol Endpoints

I/O Filters

Networking

Virtual switches

VMkernel adapters

Physical adapters

Add Networking...
Refresh
Edit...

Device	Actual Speed	Configured Speed	Switch	MAC Address	Observed I
vmnic0	10000 Mb	10000 Mb	vSwitch0	00:0c:29:37:24:be	No networ
vmnic1	10000 Mb	10000 Mb	--	00:0c:29:37:24:c8	No networ

- Add VMKernel Network for Vmotion

192.168.1.26 - Add Networking

1 Select connection type

2 Select target device

3 Port properties

4 IPv4 settings

5 Ready to complete

Select connection type

Select a connection type to create.

☒ **VMkernel Network Adapter**

The VMkernel TCP/IP stack handles traffic for ESXi services such as vSphere vMotion, iSCSI, NFS, FCoE, Fault Tolerance, vSAN and host management.

☐ **Virtual Machine Port Group for a Standard Switch**

A port group handles the virtual machine traffic on standard switch.

☐ **Physical Network Adapter**

A physical network adapter handles the network traffic to other hosts on the network.

192.168.1.26 - Add Networking

✓ 1 Select connection type

2 Select target device

3 Create a Standard Switch

4 Port properties

5 IPv4 settings

6 Ready to complete

Select target device

Select a target device for the new connection.

☐ **Select an existing network**

BROWSE ...

☐ **Select an existing standard switch**

BROWSE ...

☒ **New standard switch**

MTU (Bytes)

1500



192.168.1.26 - Add Networking

- 1 Select connection type
- 2 Select target device
- 3 Create a Standard Switch**
- 4 Port properties
- 5 IPv4 settings
- 6 Ready to complete

Create a Standard Switch

Assign free physical network adapters to the new switch.

Assigned adapters



Active adapters
 (New) vmnic1
Standby adapters
Unused adapters

All	Properties	CDP	LLDP
Adapter			VMware Inc. vmxnet3 Controller
Name			vmnic1
Location			PCI 0000:13:00.0
Driver			nvmxnet3
Status			
Status			Connected
Actual speed, Duplex			10000 Mb, Full Dupl
Configured speed, Duplex			10000 Mb, Full Dupl
Networks			No networks

192.168.1.26 - Add Networking

- 1 Select connection type
- 2 Select target device
- 3 Create a Standard Switch
- 4 Port properties**
- 5 IPv4 settings
- 6 Ready to complete

Port properties

Specify VMkernel port settings.

VMkernel port settings

Network label	Migration	
VLAN ID	None (0)	▼
MTU	Get MTU from switch ▼	1500
TCP/IP stack	Default ▼	

Available services

Enabled services

- ☒ vMotion
- ☐ Provisioning
- ☒ Fault Tolerance logging
- ☐ Management

192.168.1.26 - Add Networking

- ✓ 1 Select connection type
- ✓ 2 Select target device
- ✓ 3 Create a Standard Switch
- ✓ 4 Port properties
- 5 IPv4 settings**
- 6 Ready to complete

IPv4 settings

Specify VMkernel IPv4 settings.

☐ Obtain IPv4 settings automatically

☒ Use static IPv4 settings

IPv4 address	192.168.1.50
Subnet mask	255.255.255.0
Default gateway	☒ Override default gateway for this adapter 192.168.1.1
DNS server addresses	192.168.76.2

192.168.1.26 - Add Networking

- ✓ 1 Select connection type
- ✓ 2 Select target device
- ✓ 3 Create a Standard Switch
- ✓ 4 Port properties
- ✓ 5 IPv4 settings
- 6 Ready to complete**

Ready to complete

Review your settings selections before finishing the wizard.

New standard switch	vSwitch1
Assigned adapters	vmnic1
Switch MTU	1500
New port group	Migration
VLAN ID	None (0)
vMotion	Enabled
Provisioning	Disabled
Fault Tolerance logging	Enabled
Management	Disabled
vSphere Replication	Enabled
vSphere Replication NFC	Disabled
vSAN	Disabled
NIC settings	
MTU	1500
TCP/IP stack	Default
IPv4 settings	
IPv4 address	192.168.1.50 (static)
Subnet mask	255.255.255.0
Default gateway for IPv4	192.168.1.1

CANCEL

BACK

FINISH

4) Storage

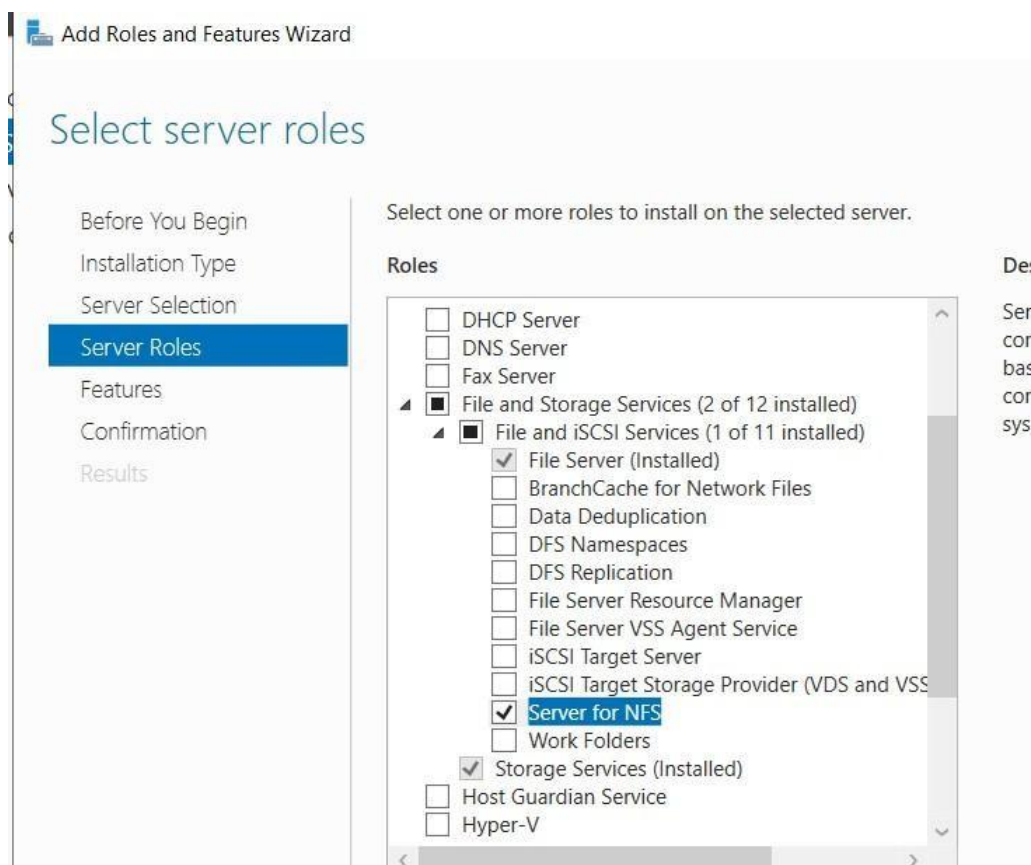
NFS (Network File System) datastore is configured to provide shared storage across the ESXi hosts. The NFS datastore resides on a Windows Server configured as an NFS server and is mounted over the

management network. By using NFS, both ESXi hosts can access the same shared storage simultaneously, enabling key vSphere features such as vMotion, High Availability (HA), Distributed Resource Scheduler (DRS), and centralized VM storage management.

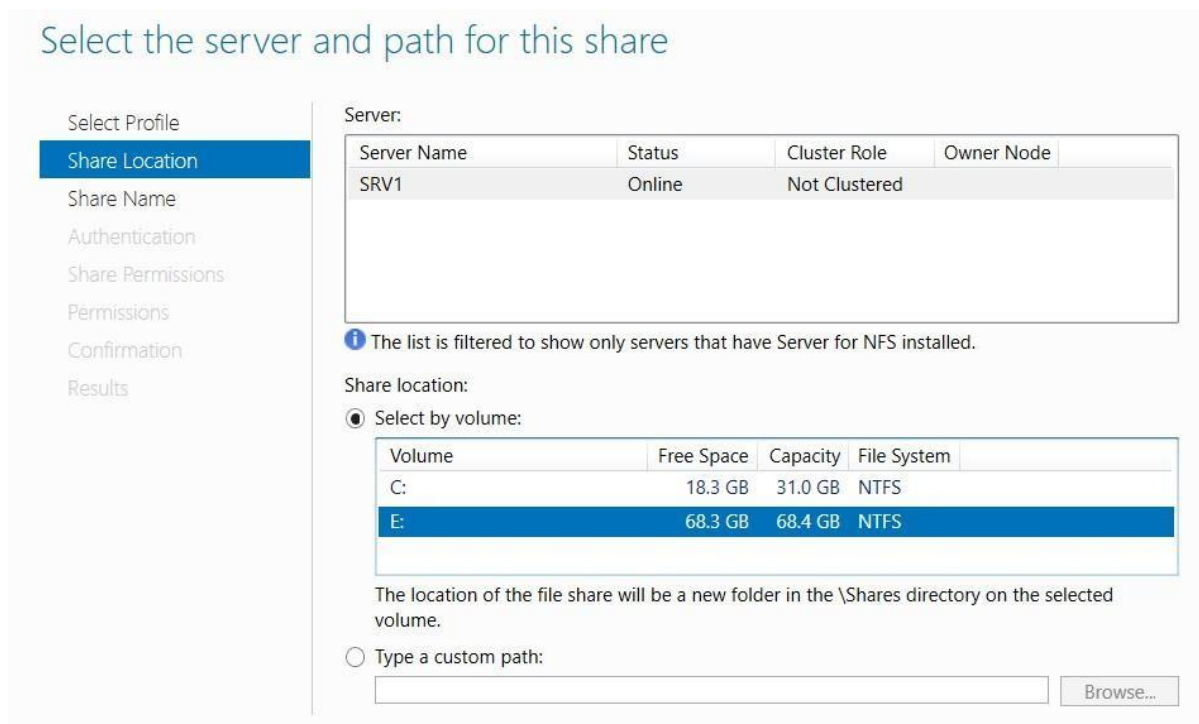
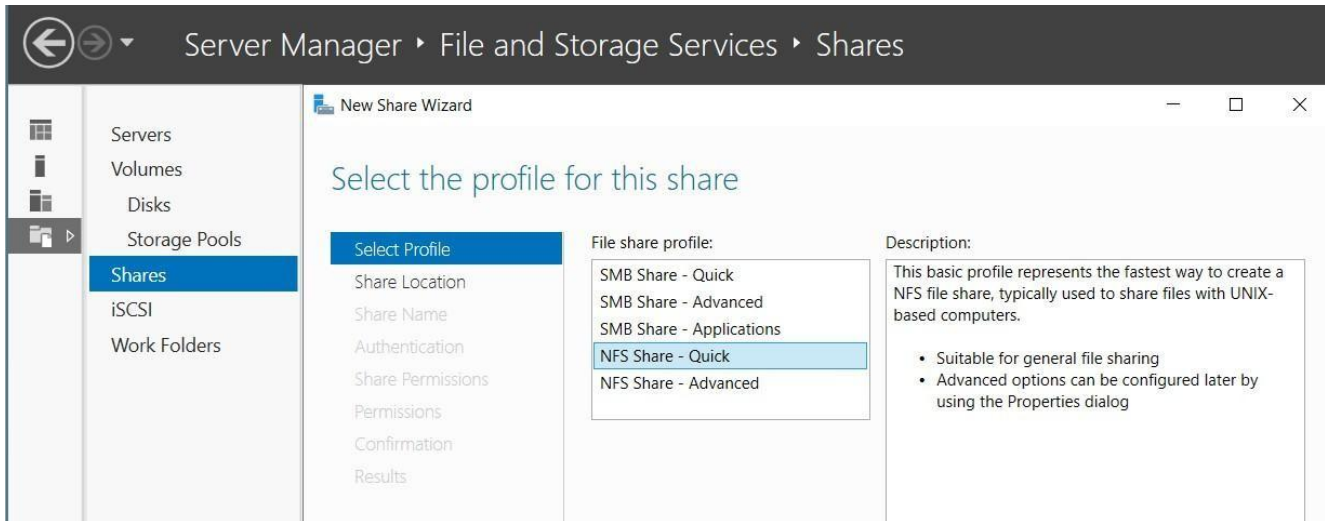
4.1) Deployment

4.1.1) Server Side (Windows Server)

To provide shared storage for the vSphere environment, an NFS server was deployed on a Windows Server.



- We used **NFS Quick Share** in Windows Server because it provides a fast and simple way to share storage over the network without requiring advanced configuration.



Specify share name

Select Profile
Share Location
Share Name
Authentication
Share Permissions
Permissions
Confirmation

Share name: Datastore_SRV1

Local path to share:

E:\Shares\Datastore_SRV1

 If the folder does not exist, the folder is created.

Remote path to share:

SRV1:/Datastore_SRV1

Specify authentication methods

Select Profile
Share Location
Share Name
Authentication
Share Permissions
Permissions
Confirmation
Results

Specify the authentication methods that you want to use for this NFS share.

☐ Kerberos v5 authentication

☐ Kerberos v5 authentication(Krb5)

☐ Kerberos v5 authentication and integrity(Krb5i)

☐ Kerberos v5 authentication and privacy(Krb5p)

☐ No server authentication

☒ No server authentication (AUTH_SYS)

☒ Enable unmapped user access

☒ Allow unmapped user access by UID/GID

☐ Allow anonymous access

Specify the share permissions

Select Profile
Share Location
Share Name
Authentication
Share Permissions
Permissions
Confirmation
Results

 Add Permissions

Grant permissions to access the NFS share to a host, client group, or netgroup.
Select the access and language encoding for the share.

☐ Host:

☐ Netgroup:

☐ Client group:

☒ All Machines

Language encoding: BIG5

Share permissions: Read / Write

the final access
share permission
ed.



Confirm selections

Select Profile
Share Location
Share Name
Authentication
Share Permissions
Permissions
Confirmation
Results

Confirm that the following are the correct settings, and then click Create.

SHARE LOCATION

Server: SRV1
Cluster role: Not Clustered
Local path: E:\Shares\Datastore_SRV1

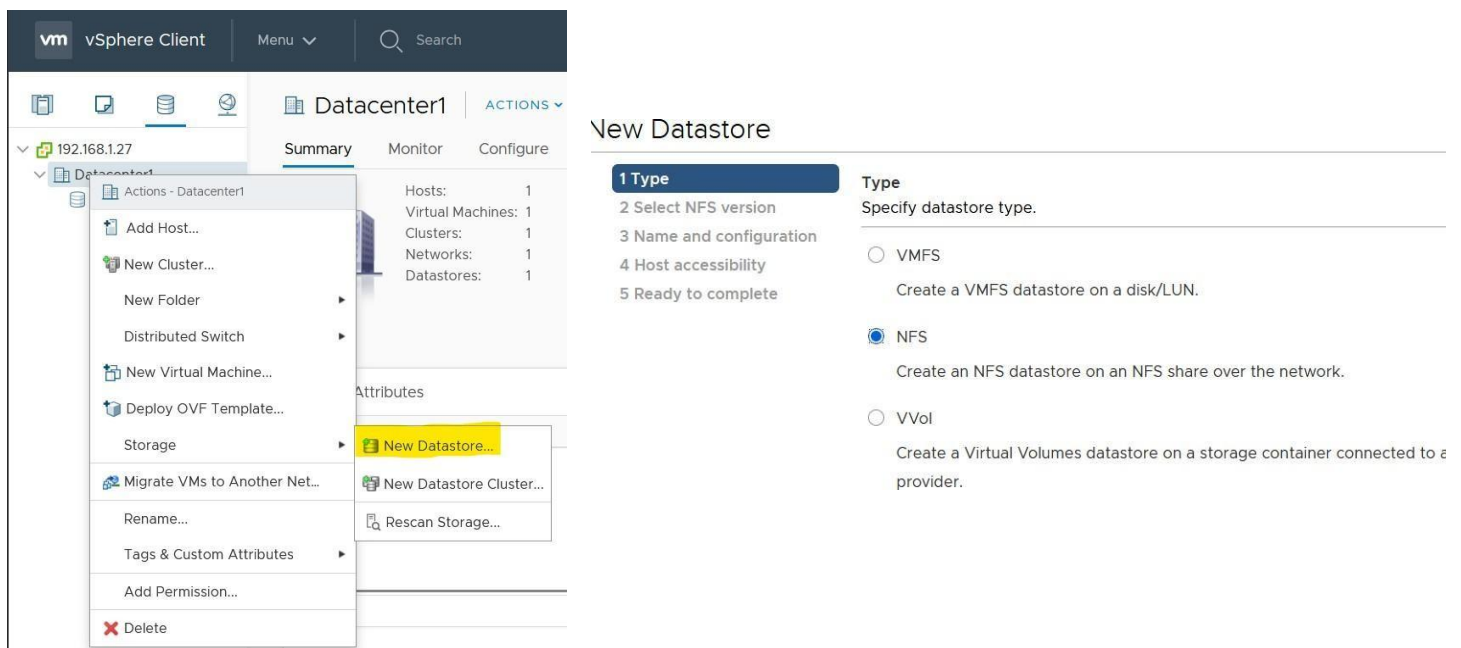
SHARE PROPERTIES

Share name: Datastore_SRV1
Protocol: NFS

4.1.2) Client Side (vCenter)

On the vCenter side, the NFS datastore is mounted to provide shared storage accessible by all ESXi hosts in the cluster.

This allows vCenter to enable advanced features like vMotion, HA, and DRS by ensuring all hosts see the same shared storage.



The screenshot shows the vSphere Client interface. On the left, the 'Datacenter1' tree is expanded, and the 'Storage' folder is selected. A context menu is open, showing the 'New Datastore...' option highlighted. The main pane displays the 'New Datastore' wizard. The 'Type' section is selected, showing three options: VMFS, NFS (selected), and VVol. The 'Summary' tab shows the configuration details for the new datastore, including the server name, cluster role, local path, share name, and protocol.

- ❑ **Authentication:** NFS v3 relies on simple AUTH_SYS (**root-based**) authentication, while NFS v4.1 introduces stronger security with optional **Kerberos** authentication.
- ❑ **Error Handling:** In NFS v3, errors are tracked on the **client side**, but in NFS v4.1, they are managed on the **server side**, improving consistency and reliability.

New Datastore

✓ 1 Type

2 Select NFS version

3 Name and configuration

4 Host accessibility

5 Ready to complete

Select NFS version

NFS Version

☒ NFS 3

NFS 3 allows the datastore to be accessed by ESX/ESXi hosts of version earlier than 6.0

☐ NFS 4.1

NFS 4.1 provides multipathing for servers and supports the Kerberos authentication protocol

- Mount the Shared Storage

New Datastore

✓ 1 Type

✓ 2 Select NFS version

3 Name and configuration

4 Host accessibility

5 Ready to complete

Name and configuration

Specify name and configuration.

ⓘ If you plan to configure an existing datastore on new hosts in the datacenter, it is recommended to use the "Mount to additional hosts" action from the datastore instead.

NFS Share Details

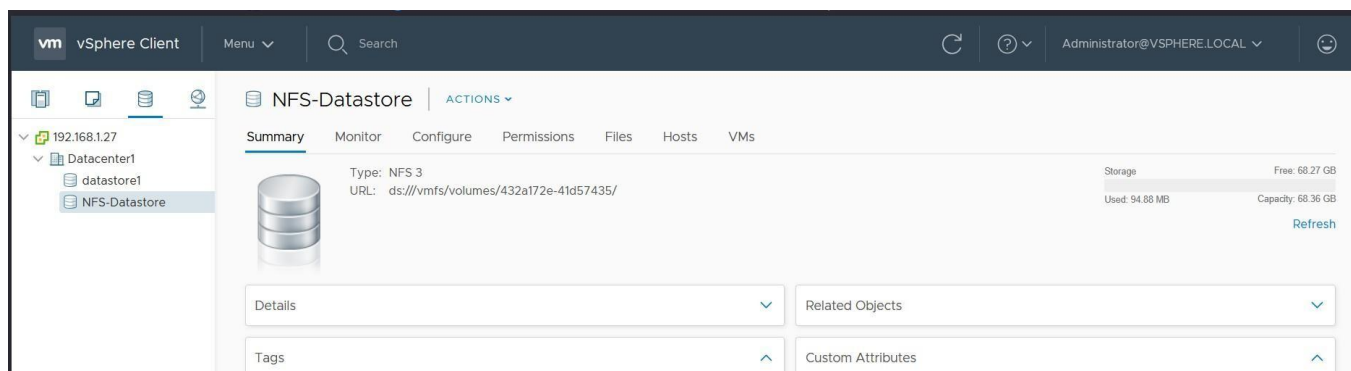
Datastore name: NFS_Datastore

Folder: /Shares

E.g: /vols/vol0/datastore-001

Server: 192.168.1.33

E.g: nas, nas.it.com or 192.168.0.1



5) Content Library

Content Library is a central repository that stores VM templates, ISOs, and scripts for consistent use across your environment. It lets you share, version, and deploy resources easily to multiple ESXi hosts and vCenter servers, improving standardization and efficiency.

5.1) Deployment

A local content library in vSphere is stored on a single vCenter Server and is only accessible to that vCenter; it's used to manage and deploy templates, ISOs, and files within the same environment.

New Content Library

1 Name and location

2 Configure content library

3 Add storage

4 Ready to complete

Name and location

Specify content library name and location.

Name: Main-Library

Notes:

Library for ISO-VM-Templates-snapshots-cloning

vCenter Server: 192.168.1.27

New Content Library

✓ 1 Name and location

2 Configure content library

3 Add storage

4 Ready to complete

Configure content library

Local libraries can be published externally and optimized for syncing over HTTP. Subscribed libraries originate from other published libraries.

☒ Local content library

New Content Library

✓ 1 Name and location

✓ 2 Configure content library

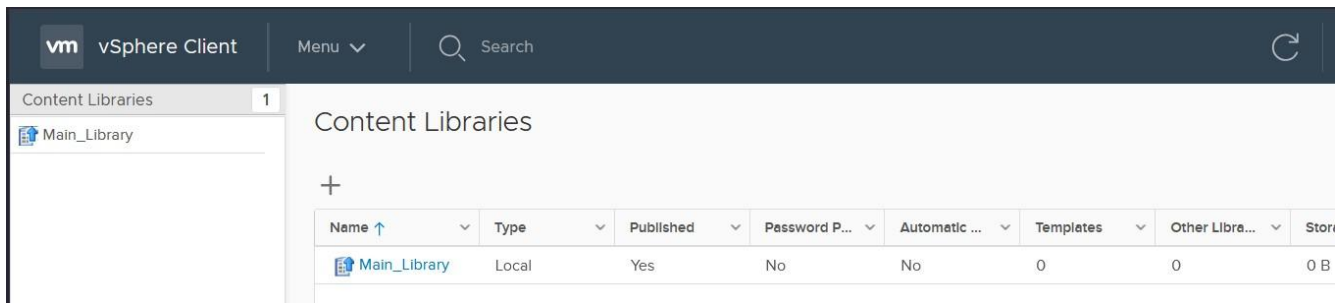
3 Add storage

4 Ready to complete

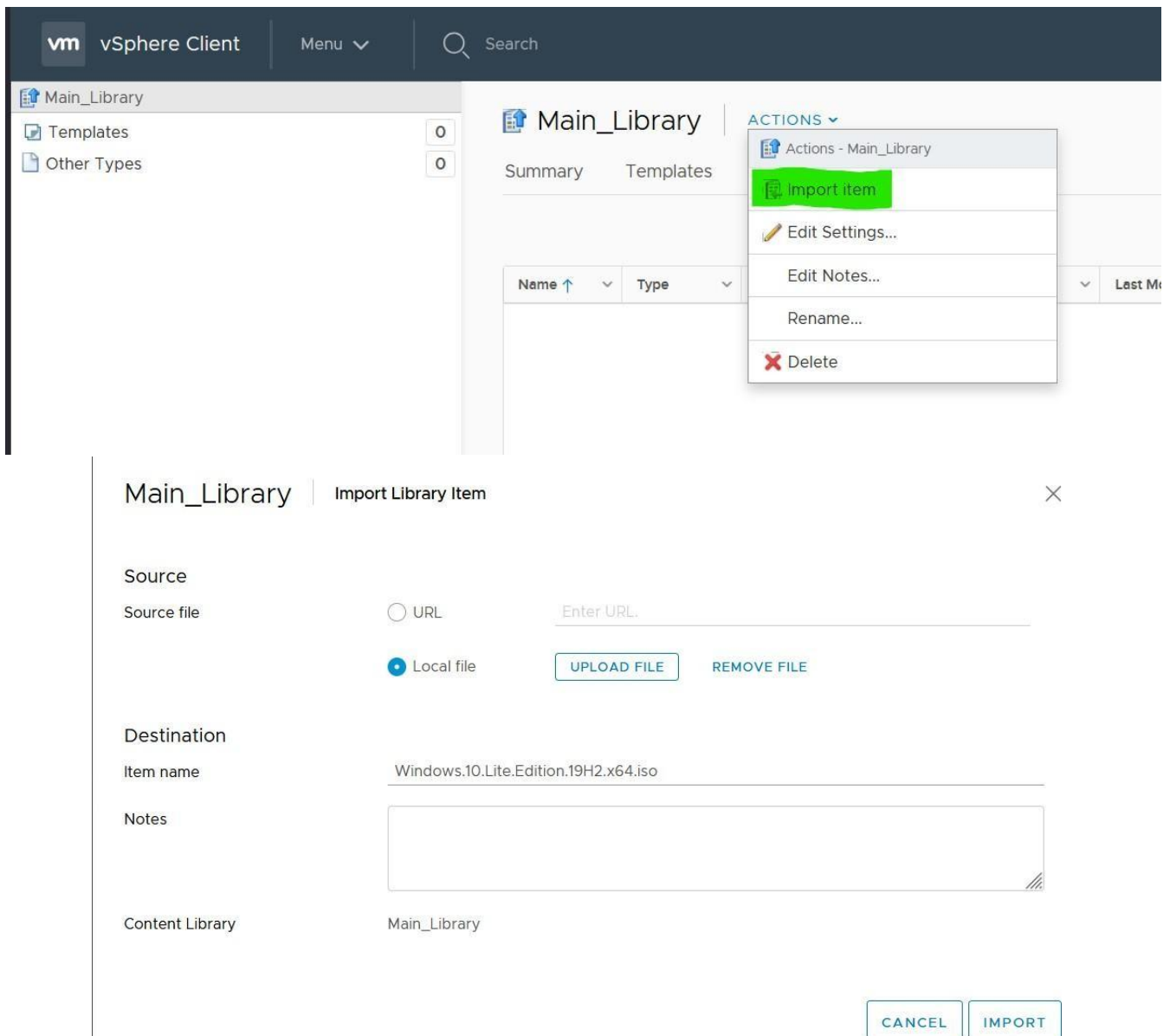
Add storage

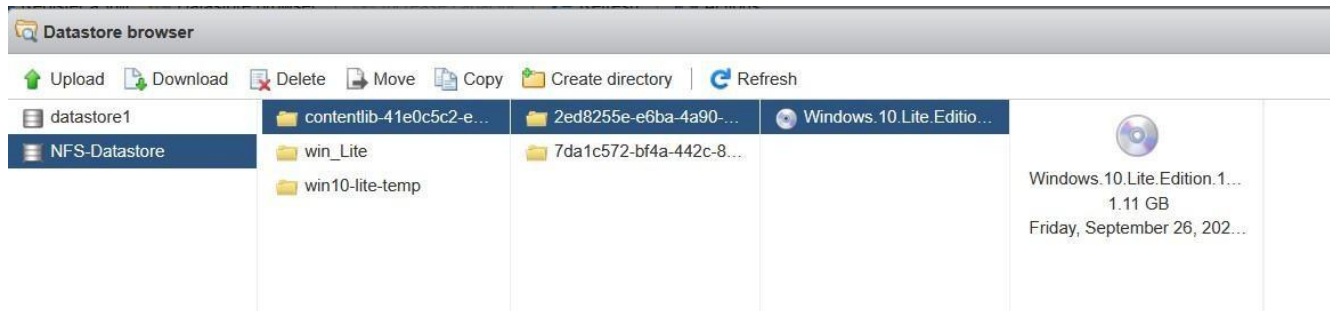
Select a storage location for the library contents. Use a file system backing for published content libraries to store the uploaded OVF packages. Use a datastore backing for local and subscribed content libraries to store content optimized for cloning.

Filter	
Name ↑	Status
datastore1	✓ Normal
NFS-Datastore	✓ Normal



5.2) Upload ISO file to Content Library





6) Creating VMs

In our project, virtual machines are created through vCenter Server to ensure centralized management and consistency. During VM creation, resources such as CPU, memory, storage, and network are allocated according to project requirements.

Ways of Creation:

VM Creation in ESXi:

A virtual machine can be created directly on an ESXi host using the ESXi web client by specifying CPU, memory, disk, and network resources, then installing an operating system from an ISO or template.

VM Creation in vCenter:

Within vCenter, VMs can be created on any connected host or cluster. vCenter provides advanced options like templates, cloning, resource pools, and policies, making VM provisioning more scalable and centralized compared to standalone ESXi.

6.1) Deployment

6.1.1) VM Standard Creation (ISO)

New Virtual Machine

1 Select a creation type

- 2 Select a name and folder
- 3 Select a compute resource
- 4 Select storage
- 5 Select compatibility
- 6 Select a guest OS
- 7 Customize hardware
- 8 Ready to complete

Select a creation type

How would you like to create a virtual machine?

- Create a new virtual machine
- Deploy from template
- Clone an existing virtual machine
- Clone virtual machine to template
- Clone template to template
- Convert template to virtual machine

This option guides you through creating a new virtual machine. You will be able to customize processors, memory, network connections, and storage. You will need to install a guest operating system after creation.

New Virtual Machine

✓ 1 Select a creation type

2 Select a name and folder

- 3 Select a compute resource
- 4 Select storage
- 5 Select compatibility
- 6 Select a guest OS
- 7 Customize hardware
- 8 Ready to complete

Select a name and folder

Specify a unique name and target location

Virtual machine name:

Select a location for the virtual machine.

- ✓ 192.168.1.27
 - Datacenter1

New Virtual Machine

✓ 1 Select a creation type

✓ 2 Select a name and folder

3 Select a compute resource

- 4 Select storage
- 5 Select compatibility
- 6 Select a guest OS
- 7 Customize hardware

Select a compute resource

Select the destination compute resource for this operation

- ✓ Datacenter1
 - ✓ ITI
 - 192.168.1.18
 - 192.168.1.26

New Virtual Machine

- ✓ 1 Select a creation type
- ✓ 2 Select a name and folder
- ✓ 3 Select a compute resource
- 4 Select storage**
- 5 Select compatibility
- 6 Select a guest OS
- 7 Customize hardware
- 8 Ready to complete

Select storage

Select the datastore in which to store the configuration and disk files

☐ Encrypt this virtual machine (Requires Key Management Server)

VM Storage Policy: Datastore Default

Name	Capacity	Provisioned	Free	Type
datastore1	142.5 GB	291.4 GB	117.38 GB	VM
NFS-Datastore	68.36 GB	1.04 GB	67.32 GB	NF

New Virtual Machine

- ✓ 1 Select a creation type
- ✓ 2 Select a name and folder
- ✓ 3 Select a compute resource
- ✓ 4 Select storage
- ✓ 5 Select compatibility
- ✓ 6 Select a guest OS
- 7 Customize hardware**
- 8 Ready to complete

Configure the virtual machine hardware

Virtual Hardware VM Options

ADD NEW DEVICE

> CPU	1		
> Memory	1	GB	
> New Hard disk *	16	GB	
> New SCSI controller *	VMware Paravirtual		
> New Network *	VM Network		☒ Connect...
> New CD/DVD Drive *	Content Library ISO File		☐ Connect...
> Video card *	Specify custom settings		
VMCI device		Device on the virtual machine PCI bus that provides support for the virtual machine communication interface	
> Other	Additional Hardware		

vm vSphere Client

Menu

Search

Administrator@VSPHERE.LOCAL

192.168.1.27

Datacenter1

Project_cluster

192.168.1.26

vm_tiny

vsphere

vm_tiny

ACTIONS

Summary Monitor Configure Permissions Datastores Networks

vmware

Powered On

Launch Web Console

Launch Remote Console

Guest OS:

Other 4.x or later Linux (64-bit)

Compatibility:

ESXi 6.7 and later (VM version 14)

VMware Tools:

Not running, not installed

DNS Name:

More info

IP Addresses:

Host: 192.168.1.26

CPU USAGE

0 Hz

MEMORY USAGE

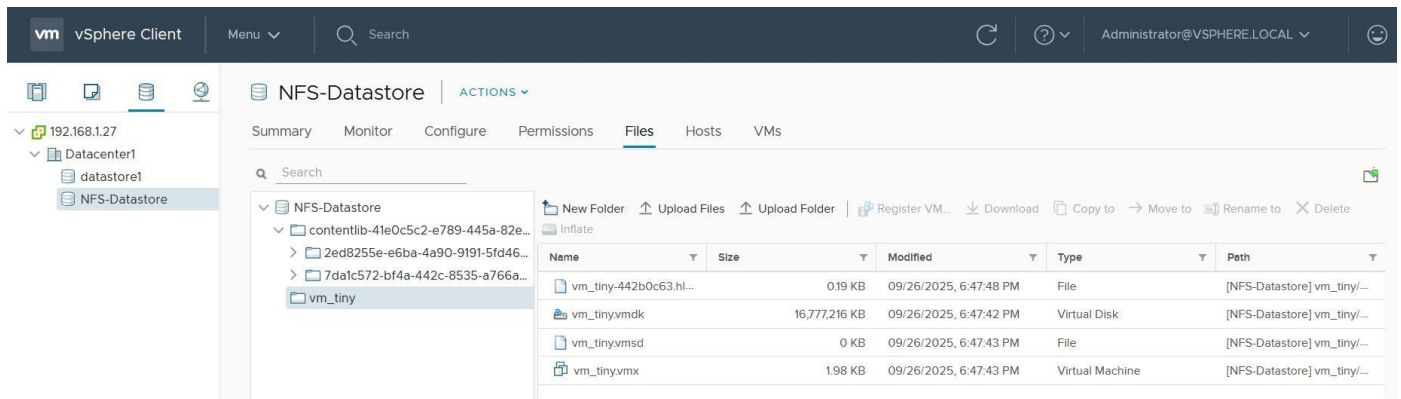
0 B

STORAGE USAGE

17.08 GB

VMware Tools is not installed on this virtual machine.

Install VMware Tools...



Ways to Create a VM in vSphere

1. **Create New VM (from scratch)**
 - Manually configure CPU, RAM, disk, and network.
 - Install OS using ISO from datastore or content library.
 - Useful for one-off or custom builds.
2. **From Template**
 - Deploy a VM from a standardized golden image (OS + base configuration).
 - Ensures consistency and saves time for repeated deployments.
 - Often stored in a content library for easy access across clusters.
3. **From Clone**
 - Make a full copy of an existing VM.
 - Can be done online (hot clone) or offline.
 - Useful when you need an exact duplicate for testing, staging, or production.
4. **From Snapshot (Indirect)**
 - Snapshots themselves are not used to “create” new VMs directly.
 - But you can revert a VM to a snapshot state and then clone it → effectively creating a new VM based on that point-in-time.
 - Useful for lab/testing scenarios.
5. **Deploy from Content Library**
 - Store templates, ISOs, and OVF files in the content library.
 - VMs can be deployed directly from these items across the environment.
 - Ensures centralized and consistent deployment.
6. **Import OVF/OVA**
 - Deploy a VM from a pre-packaged appliance or exported VM.

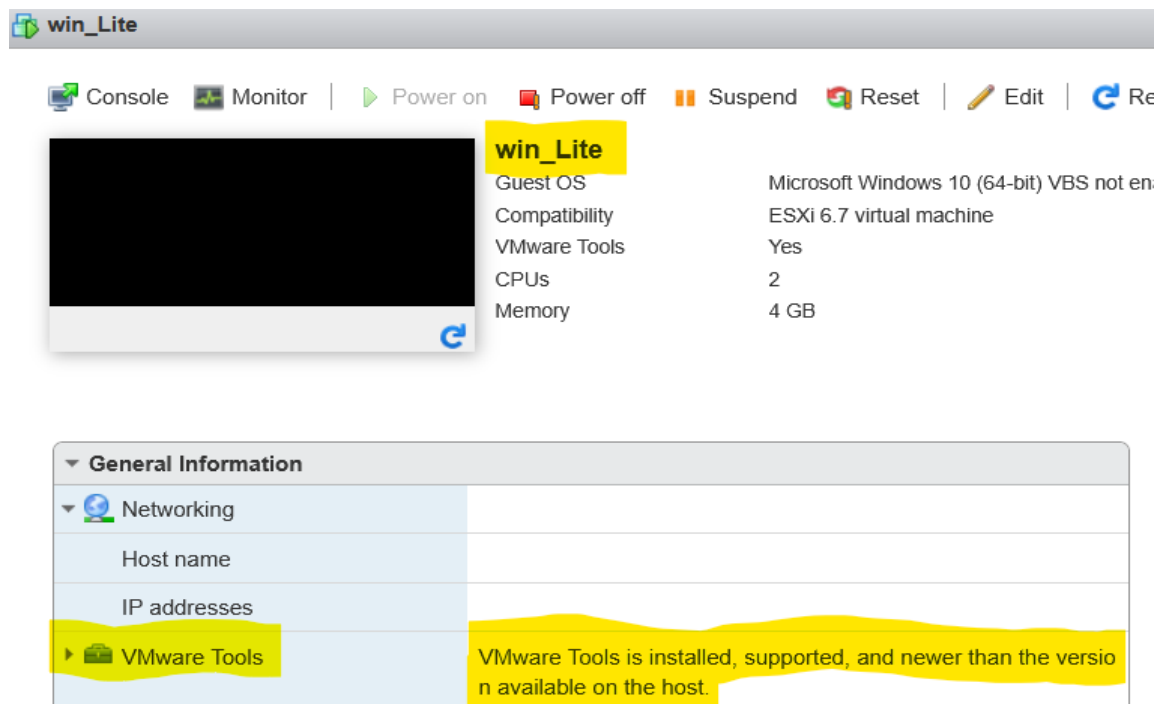
- Often used for vendor-provided appliances (firewalls, monitoring tools, etc.)

6.2) Install VMWare Tools

VMware Tools enhances VM performance and manageability by enabling features like optimized graphics, time synchronization, better network and disk performance, and graceful guest OS shutdown/restart from vCenter. It is essential for smooth monitoring, management, and integration of VMs within the vSphere environment.

- Tiny Linux:

Tiny Linux is a lightweight Linux distribution designed to run with minimal CPU, memory, and storage requirements. We use it in our environment because it consumes very low resources, making it ideal for testing and running in limited-capacity infrastructure, but it doesn't support VMware Tools so we decided to create another VM (windows 10 Lite) which supports installing VMware Tools



In our windows VM, we installed VMware tools as the following steps:

- Log in to vCenter and open the VM console.
- From the VM menu, select Guest OS → Install/Upgrade VMware Tools.
- This mounts the VMware Tools ISO to the VM's virtual CD/DVD drive.
- Inside the VM, open the CD/DVD drive and run the installer (for Windows: setup.exe, for Linux: mount and run the installer script).

- **Follow the on-screen instructions to complete installation.**
- **Reboot the VM (if prompted) to apply changes.**

7) Templates - Snapshots - Cloning

1. Template

A golden image of a VM used for deploying new VMs quickly and consistently.

Cannot be powered on (converted back to VM if needed).

Ensures standardization across the environment.

Best used when you want to deploy multiple identical VMs (like OS + base software).

2. Snapshot

A point-in-time state of a VM (disk + memory + settings).

Used for testing, or rollback before changes/patches.

Not suitable for long-term use (can affect performance and consume storage).

Best used when you need to revert back after a risky operation.

3. Clone

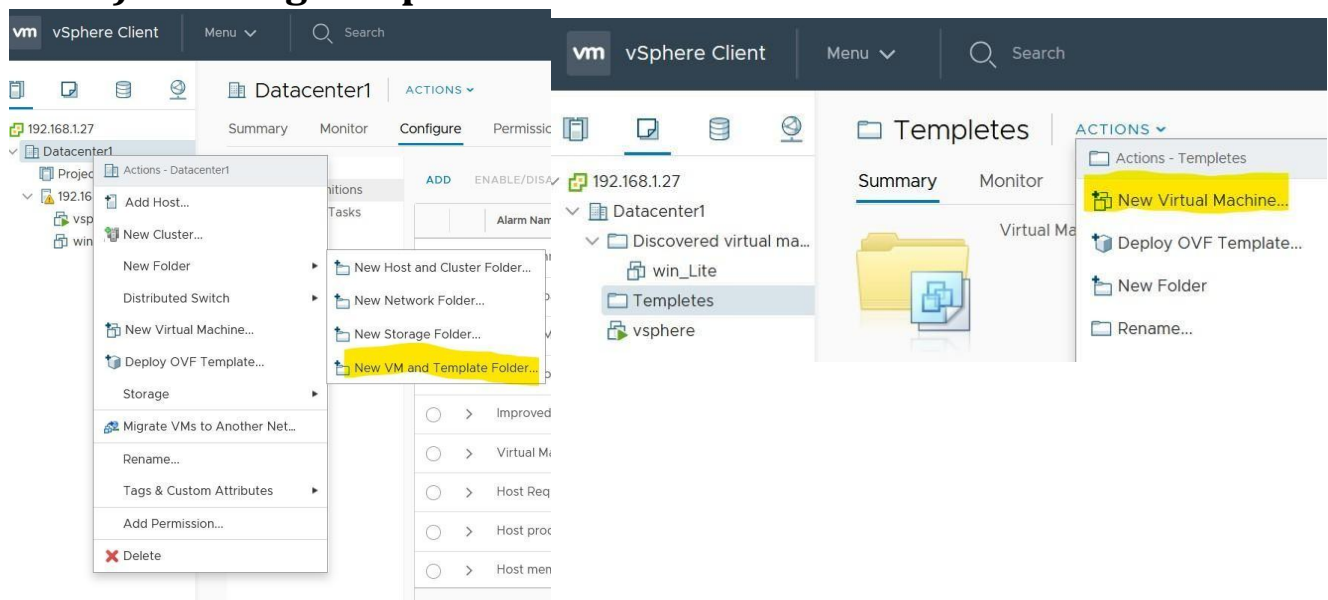
A full copy of a VM at the time of cloning.

Independent from the parent VM (can be powered on immediately). Can

be Full Clone (complete copy, large size) or Linked Clone (shares disk, smaller size).

Best used when you need another VM right away with the same configuration.

7.1) Creating Template



Clone Virtual Machine To Template

1 Select a creation type

- 2 Select a virtual machine
- 3 Select a name and folder
- 4 Select a compute resource
- 5 Select storage
- 6 Ready to complete

Select a creation type

How would you like to create a virtual machine?

- Create a new virtual machine
- Deploy from template
- Clone an existing virtual machine
- Clone virtual machine to template
- Clone template to template
- Convert template to virtual machine

Clone Virtual Machine To Template

1 Select a creation type

2 Select a virtual machine

- 3 Select a name and folder
- 4 Select a compute resource
- 5 Select storage
- 6 Ready to complete

Select a virtual machine

Select a virtual machine to clone

- 192.168.1.27
 - Datacenter1
 - Discovered virtual machine
 - win_Lite
 - Templates
 - vsphere

win_Lite - Clone Virtual Machine To Template

1 Select a creation type

2 Select a virtual machine

3 Select a name and folder

- 4 Select a compute resource
- 5 Select storage
- 6 Ready to complete

Select a name and folder

Specify a unique name and target location

VM template: win10-lite-temp
name:

Select a location for the template.

- 192.168.1.27
 - Datacenter1
 - Discovered virtual machine
 - Templates

win_Lite - Clone Virtual Machine To Template

1 Select a creation type

2 Select a virtual machine

3 Select a name and folder

4 Select a compute resource

5 Select storage

- 6 Ready to complete

Select storage

Select the datastore in which to store the configuration and disk files

Configure per disk

Select virtual disk format: Thin Provision

VM Storage Policy: Keep existing VM storage policies

Name	Capacity	Provisioned	Free
Storage Compatibility: Compatible			
datastore1	142.5 GB	301.1 GB	107.48 GB
NFS-Datastore	68.36 GB	21.35 GB	51.25 GB

win_Lite - Clone Virtual Machine To Template

1 Select a creation type

2 Select a virtual machine

3 Select a name and folder

4 Select a compute resource

5 Select storage

6 Ready to complete

Ready to complete

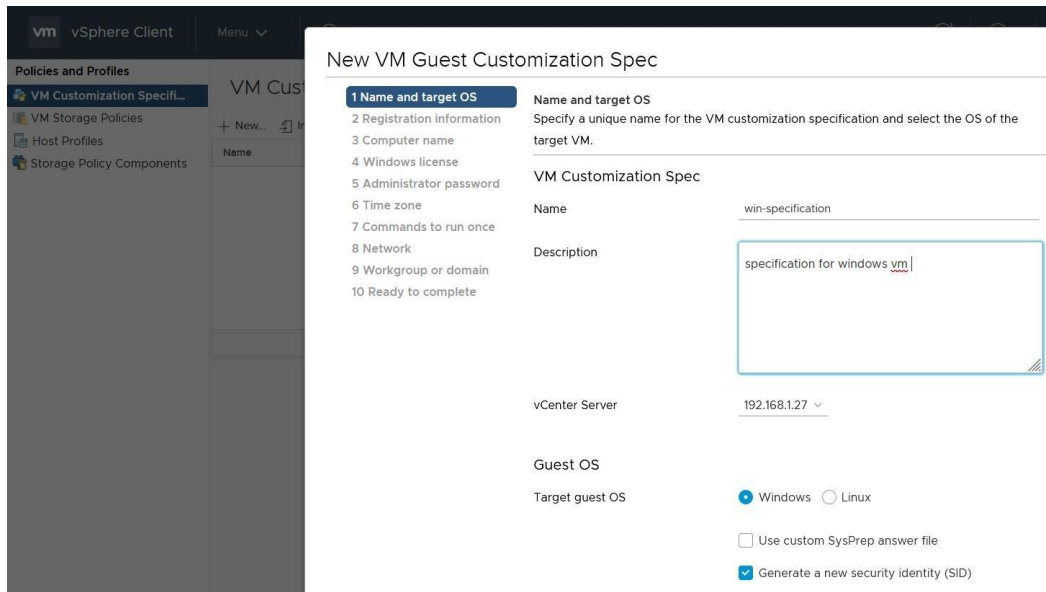
Click Finish to start creation.

Provisioning type	Clone virtual machine to template
Source virtual machine	win_Lite
Template name	win10-lite-temp
Folder	Templates
Host	192.168.1.26
Datastore	NFS-Datastore
Disk storage	Thin Provision

7.1.1) VM Customization Specification

The VM Customization Specification in vSphere allows you to automatically adjust system settings when deploying a VM from a template. It can define the computer name, domain membership, network settings (IP, DNS, gateway), and license or SID for Windows. This ensures each deployed VM is unique while still based on the same standardized template. Customization specifications save time, reduce manual errors, and provide consistency across deployments.

- From Menu tab --> Select Policies and Profiles



New VM Guest Customization Spec

1 Name and target OS
2 Registration information
3 Computer name
4 Windows license
5 Administrator password
6 Time zone
7 Commands to run once
8 Network
9 Workgroup or domain
10 Ready to complete

Name and target OS
Specify a unique name for the VM customization specification and select the OS of the target VM.

VM Customization Spec

Name: win-specification

Description: specification for windows vm

vCenter Server: 192.168.1.27

Guest OS: Windows (selected), Linux

Target guest OS: ☒ Generate a new security identity (SID)

New VM Guest Customization Spec

✓ 1 Name and target OS
✓ 2 Registration information
3 Computer name
4 Windows license
5 Administrator password
6 Time zone
7 Commands to run once
8 Network
9 Workgroup or domain
10 Ready to complete

Computer name
Specify a computer name that will identify this virtual machine on a network.

☒ Use the virtual machine name
☐ Enter a name in the Clone/Deploy wizard
☐ Enter a name

☐ Append a numeric value

New VM Guest Customization Spec

✓ 1 Name and target OS
✓ 2 Registration information
✓ 3 Computer name
✓ 4 Windows license
5 Administrator password
6 Time zone
7 Commands to run once
8 Network
9 Workgroup or domain
10 Ready to complete

Administrator password
Enter the password and auto logon option for the administrator account.

Password:

Confirm password:

☐ Automatically logon as Administrator

Number of times to logon automatically: 1

Edit Network
NIC1

IPv4
IPv6
DNS
WINS

Settings

☐ Use DHCP to obtain an IPv4 address automatically.
☒ Prompt the user for an IPv4 address when the specification is used
☐ Use an application configured on the vCenter Server to generate an IP address
Argument
☐ Use custom settings
IPv4 address

Subnet and Gateways ⓘ

Subnet mask
Default gateway
Alternate gateway

New VM Guest Customization Spec

✓ 1 Name and target OS
✓ 2 Registration information
✓ 3 Computer name
✓ 4 Windows license
✓ 5 Administrator password
✓ 6 Time zone
✓ 7 Commands to run once
8 Network
9 Workgroup or domain
10 Ready to complete

Network
Specify the network settings for the virtual machine.
☐ Use standard network settings for the guest operating system, including enabling DHCP on all network interfaces
☒ Manually select custom settings

ADD

Description	IPv4 Address	IPv6 Address
NIC1	Prompt user	Not used

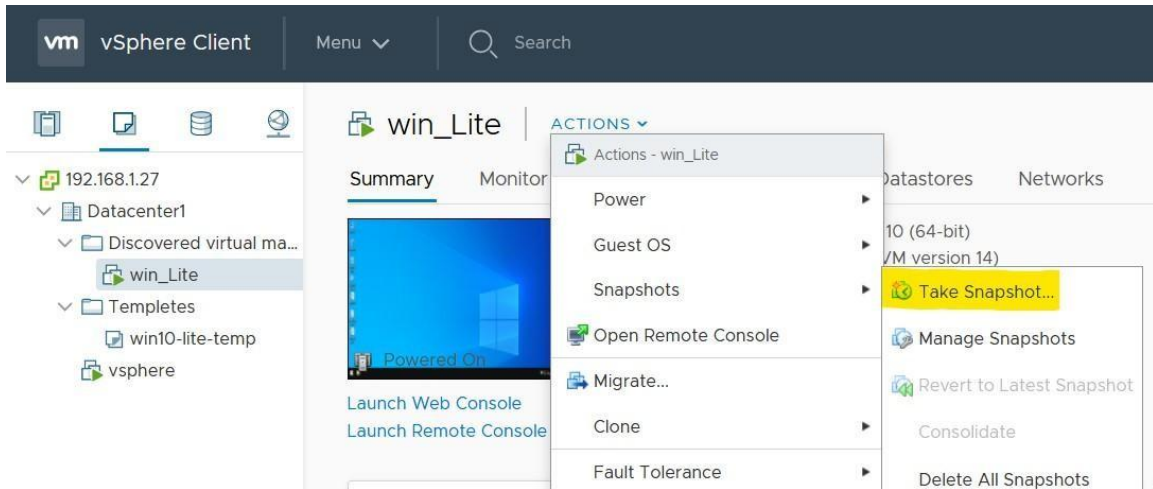
New VM Guest Customization Spec

✓ 1 Name and target OS
✓ 2 Registration information
✓ 3 Computer name
✓ 4 Windows license
✓ 5 Administrator password
✓ 6 Time zone
✓ 7 Commands to run once
✓ 8 Network
✓ 9 Workgroup or domain
10 Ready to complete

Ready to complete
Review your settings selections before finishing the wizard.

Name	win-specification
OS type	Windows
OS options	Generate new security ID
Registration info	Owner name: mostafa Organization: nader
Computer name	Use Virtual Machine name
Product key	No product key specified
Server license mode	Per Server (Maximum Connections: 5)
Administrator log in	Do not log in automatically as Administrator
Time zone	(GMT+03:00) Kuwait, Riyadh
Network type	Custom
NIC1 IPv4	Prompt user
NIC1 IPv6	Not used
Workgroup/Domain	Workgroup: WORKGROUP

7.2) Taking Snapshot



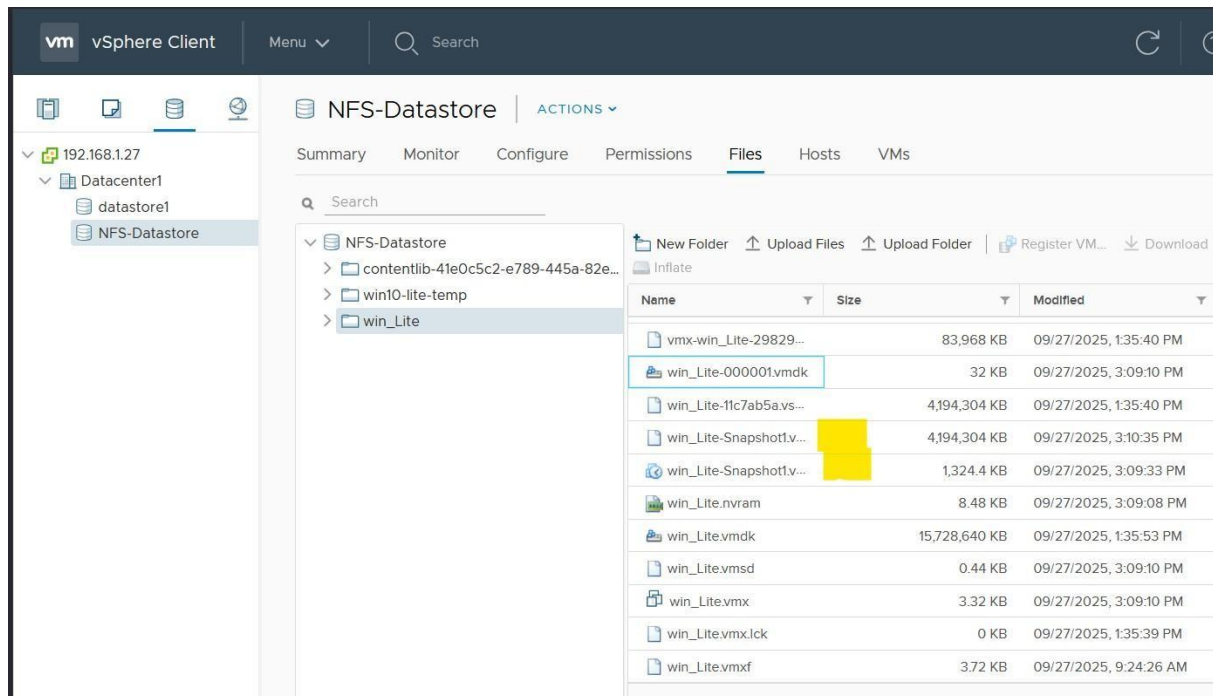
Take Snapshot | win_Lite

Name

Description

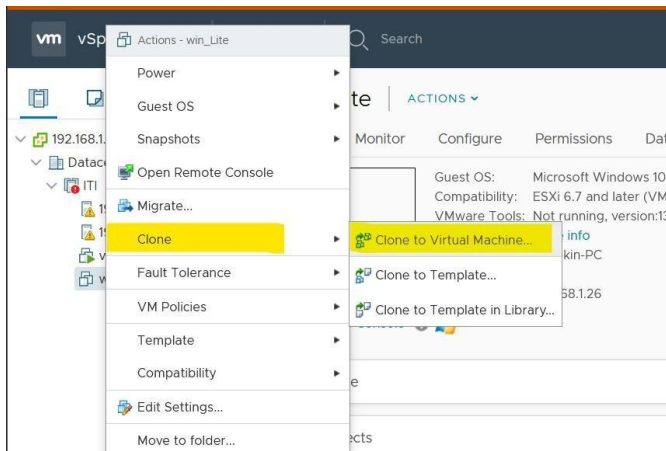
after install firefox

- ☒ Snapshot the virtual machine's memory
- ☐ Quiesce guest file system (Needs VMware Tools installed)



7.3) Clone

In VMware vSphere, cloning refers to the process of creating an exact copy of a virtual machine (VM) or virtual disk. This operation is typically used to rapidly deploy new VMs from an existing template or to replicate an existing VM for testing, backup, or scaling purposes. Cloning is beneficial in environments using VMware vCenter and ESXi hosts, as it allows for the duplication of configurations, OS settings, and installed applications.



win_Lite - Clone Existing Virtual Machine

1 Select a name and folder

2 Select a compute resource

3 Select storage

4 Select clone options

5 Ready to complete

Select a name and folder

Specify a unique name and target location

Virtual machine name: win_Lite_Clone

Select a location for the virtual machine.

192.168.1.100

Datacenter1

win_Lite - Clone Existing Virtual Machine

✓ 1 Select a name and folder

2 Select a compute resource

3 Select storage

4 Select clone options

5 Ready to complete

Select a compute resource

Select the destination compute resource for this operation

Datacenter1

IT1

192.168.1.18

192.168.1.26

win_Lite - Clone Existing Virtual Machine

✓ 1 Select a name and folder

✓ 2 Select a compute resource

3 Select storage

4 Select clone options

5 Ready to complete

Select storage

Select the datastore in which to store the configuration and disk files

Configure per disk ☐

Select virtual disk format: Thin Provision

VM Storage Policy: Keep existing VM storage policies

Name	Capacity	Provisioned	Free
Storage Compatibility: Compatible			
datastore1 (1)	52.5 GB	1.41 GB	51.09 GB
NFS_Datastore	68.36 GB	36.41 GB	31.95 GB

- While applying the VM cloning you need to specify which VM Customization Specification you will use to specify option like network settings.

win_Lite - Clone Existing Virtual Machine

- ✓ 1 Select a name and folder

✓ 2 Select a compute resource

✓ 3 Select storage

4 Select clone options

5 Customize guest OS

6 Ready to complete
- Select clone options

Select further clone options

☒ Customize the operating system

☐ Customize this virtual machine's hardware (Experimental)

☐ Power on virtual machine after creation

win_Lite - Clone Existing Virtual Machine

- ✓ 1 Select a name and folder

✓ 2 Select a compute resource

✓ 3 Select storage

✓ 4 Select clone options

5 Customize guest OS

6 Ready to complete
- Customize guest OS

Customize the guest OS to prevent conflicts when you deploy the virtual machine

Operating System: Microsoft Windows 10 (64-bit)

Name ↑	Guest OS	Last Modified
win-specification	Windows	09/29/2025, 10:14:45 PM

win_Lite - Clone Existing Virtual Machine

- ✓ 1 Select a name and folder

✓ 2 Select a compute resource

✓ 3 Select storage

✓ 4 Select clone options

✓ 5 Customize guest OS

6 User settings

7 Ready to complete
- User settings

To customize your user specification, specify the required settings for the virtual machine.

Network Adapter Settings *

Network Adapter 1

IPv4 address *	192.168.1.23
IPv4 subnet mask *	255.255.255.0
IPv4 default gateway	192.168.1.1
IPv4 alternate gateway	192.168.1.18

win_Lite - Clone Existing Virtual Machine

- ✓ 1 Select a name and folder

✓ 2 Select a compute resource

✓ 3 Select storage

✓ 4 Select clone options

✓ 5 Customize guest OS

✓ 6 User settings

7 Ready to complete
- Ready to complete

Click Finish to start creation.

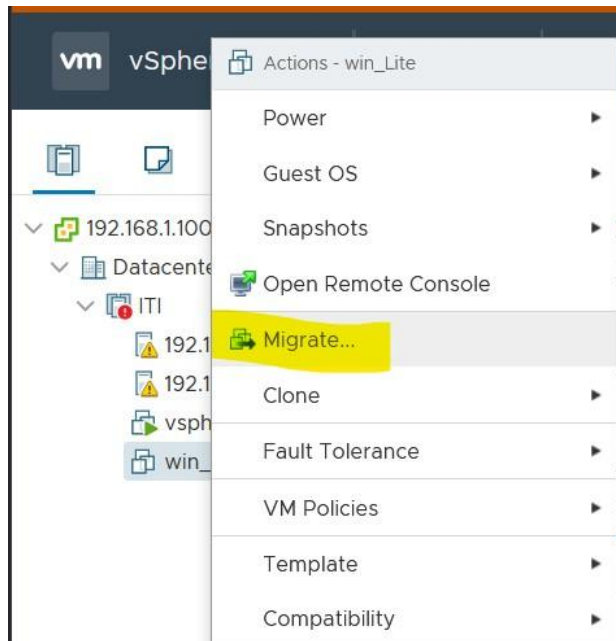
Provisioning type	Clone an existing virtual machine
Source virtual machine	win_Lite
Virtual machine name	win_Lite_clone
Folder	Datacenter1
Host	192.168.1.18
Datastore	NFS_Datastore
Disk storage	Thin Provision
Guest OS customization specification	win-specification

8) Migration (vMotion)

VM Migration in VMware ESXi refers to the process of moving a virtual machine (VM) from one host to another without downtime.

This is essential for resource optimization, load balancing, and high availability in virtualized environments.

- While migration you can specify migration type, we choose to migrate both resource Compute and storage.



win_Lite - Migrate

1 Select a migration type

- 2 Select a compute resource
- 3 Select storage
- 4 Select networks
- 5 Ready to complete

Select a migration type

Change the virtual machines' compute resource, storage, or both.

☐ **Change compute resource only**

Migrate the virtual machines to another host or cluster.

☐ **Change storage only**

Migrate the virtual machines' storage to a compatible datastore.

☒ **Change both compute resource and storage**

Migrate the virtual machines to a specific host or cluster and to a specific datastore or datastore cluster.

win_Lite - Migrate

✓ 1 Select a migration type

2 Select a compute resource

3 Select storage

4 Select networks

5 Ready to complete

Select a compute resource

Select a cluster, host, vApp or resource pool to run the virtual machines.

✓ 192.168.1.100

✓ Datacenter1

✓ IT1

192.168.1.18

192.168.1.26

win_Lite - Migrate

- ✓ 1 Select a migration type
- ✓ 2 Select a compute resource
- 3 Select storage**
- 4 Select networks
- 5 Ready to complete

Select storage

Select the destination storage for the virtual machine migration.

Configure per disk ☐

Select virtual disk format: Thin Provision

VM Storage Policy: Keep existing VM storage policies

Name	Capacity	Provisioned	Free	T
Storage Compatibility: Compatible				
NFS_Datastore	68.36 GB	36.41 GB	31.95 GB	
datastore1 (1)	52.5 GB	1.41 GB	51.09 GB	

win_Lite - Migrate

- ✓ 1 Select a migration type
- ✓ 2 Select a compute resource
- ✓ 3 Select storage
- 4 Select networks**
- 5 Ready to complete

Select networks

Select destination networks for the virtual machine migration.

Migrate VM networking by selecting a new destination network for all VM network adapters attached to the same source network.

Source Network	Used By	Destination Network
VM Network	1 VMs / 1 Network adapters	Move

VM Network is in use at:

VM	Network Adapter	Network
win_Lite	Network adapter 1	VM Network

win_Lite - Migrate

- ✓ 1 Select a migration type
- ✓ 2 Select a compute resource
- ✓ 3 Select storage
- ✓ 4 Select networks
- 5 Ready to complete**

Ready to complete

Verify that the information is correct and click Finish to start the migration.

Migration Type	Change compute resource and storage
Virtual Machine	win_Lite
Cluster	[IT]
Host	192.168.1.18
Storage	NFS_Datastore
Disk Format	Thin Provision
Networks	Virtual network adapters from VM Network will be reassigned to "Move"

vm vSphere Client

Menu Search Administrator@VSPHERE.LOCAL

NFS_Datastore ACTIONS

Summary Monitor Configure Permissions Files Hosts VMs

Search

NFS_Datastore

- > vSphere-HA
- > contentlib-41e0c5c2-e789-445a-82e...
- > win10-lite-temp
- > win_Lite
- > win_Lite_1

New Folder Upload Files Upload Folder Register VM... Download Copy to Move to Rename to Delete

Inflate

Name	Size	Modified	Type	Path
win_Lite-279ae7a4-hl...	0.92 KB	09/29/2025, 10:37:33 PM	File	[NFS_Datastore] win_Lite...
win_Lite.vmdk	15,728,640 KB	09/29/2025, 10:37:44 PM	Virtual Disk	[NFS_Datastore] win_Lite...

9) HA & DRS

High Availability (HA) in vSphere automatically restarts virtual machines on healthy hosts if a host fails, ensuring minimal downtime. Distributed Resource Scheduler (DRS) balances workloads across hosts in a cluster by automatically migrating VMs (vMotion) to optimize performance and resource utilization.

9.1) DRS Modes:

Manual

Administrator must approve and trigger the move. Useful if you want full control.

Partially Automated

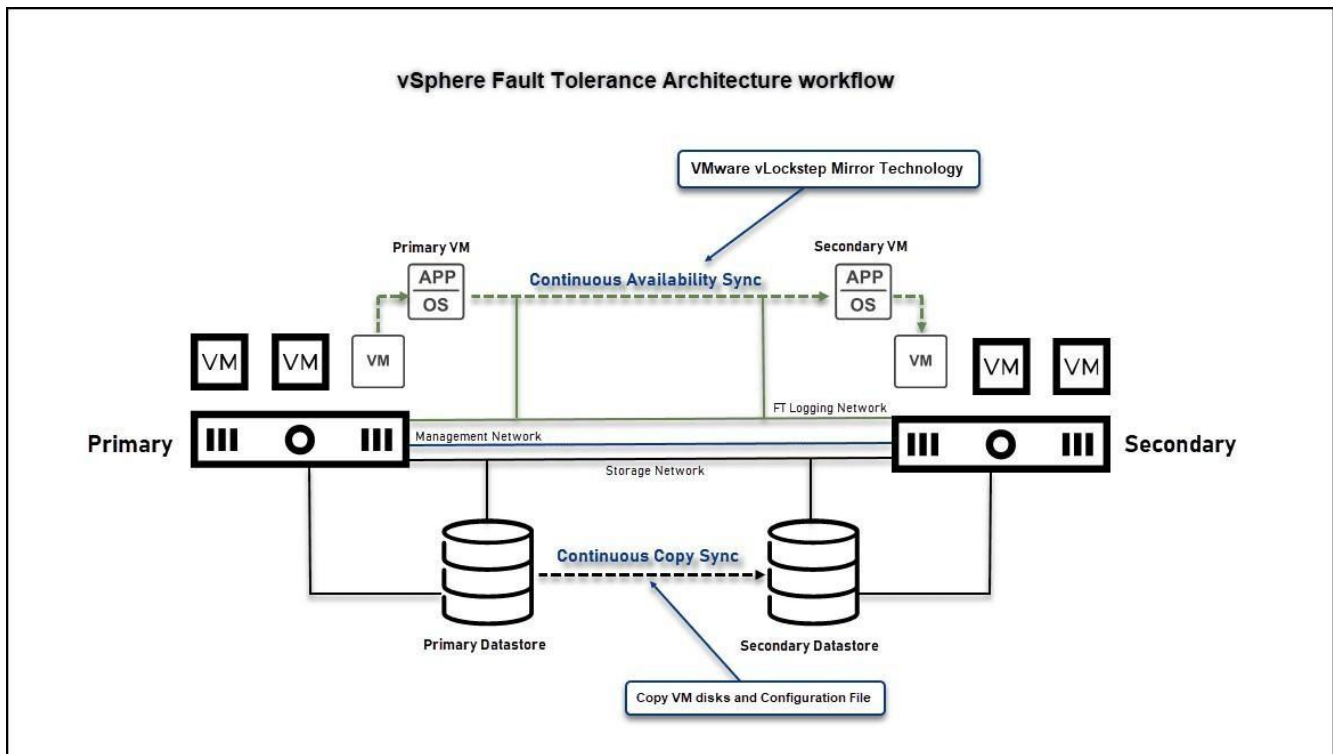
At VM power-on: DRS automatically decides the best host to start the VM. During runtime: It will only recommend migrations, not perform them automatically.

Fully Automated

DRS places VMs on the best host at power-on. Continuously monitors cluster balance and migrates VMs automatically using vMotion when needed.

10) Fault Tolerance

In our vSphere project, Fault Tolerance (FT) is a high-availability feature that ensures continuous availability for virtual machines. It works by creating a live shadow copy of a VM (the secondary VM) on another ESXi host within the cluster. If the primary host fails, the secondary VM immediately takes over with no downtime or data loss. This setup builds on vMotion and shared storage (like our NFS datastore) to keep both VMs synchronized in real-time, ensuring business continuity even during host failures.



11) Project Summary

The vSphere Project aims to design and implement a fully functional virtualized data center environment using VMware vSphere. The project demonstrates enterprise-level virtualization capabilities by integrating multiple ESXi hosts, a vCenter Server, and shared storage to enable advanced features such as vMotion, High Availability (HA), and Fault Tolerance (FT).

The infrastructure consists of two ESXi hosts deployed within VMware Workstation, centrally managed by a vCenter Server Appliance (VCSA). The environment is connected through a shared NFS datastore, which allows both hosts to access common storage for virtual machine files, templates, and ISOs. The NFS datastore is set up using a Windows Server configured as the storage provider, ensuring centralized data access and simplified management.

The vCenter Server serves as the control point for managing and monitoring the virtual infrastructure. It enables the creation of clusters that combine multiple ESXi hosts to balance workloads, ensure resource optimization, and provide high availability. Features such as Distributed Resource Scheduler (DRS) and vMotion are enabled within the cluster to allow seamless migration of VMs between hosts without downtime, improving performance and availability.

High Availability (HA) is configured to automatically restart virtual machines on healthy hosts in case of host failure, minimizing service interruption. Fault Tolerance (FT) provides an additional layer of protection by running an identical secondary VM in lockstep with the primary VM, guaranteeing continuous service even if one host fails.

The project also includes a Content Library, which stores VM templates and ISO images to streamline the deployment and management of virtual machines. Additionally, various test VMs, including lightweight operating systems like win10 lite , are deployed to simulate workloads and validate the environment's performance and functionality.